

local test

Simultaneous Equations

1. Solve the simultaneous equations: $4x^2 + 3xy + y^2 = 8$ and $xy + 4 = 0$ [6 marks]

2. Solve the following simultaneous equations: $e^{(x+y)} \times e^{(3x-2y)} = 1$ and $x^{2y} = 256$. [5 marks]

3. Curves $4x^2 - 3y^2 + xy = 24$ and $y = 2/x$ intersect at P and Q. Find coordinates. [5 marks]

4. Hence find the exact solutions of the following simultaneous equations. $256^{(x+y)} \times 16^{(-2x)} = 8^{(-x+3y)}$, $x^2 + 3y^2 = 56$ [3 marks]

5. Find the coordinates of the points where the curve with equation $2x^2 + 8xy = x^3y - 20x$ intersects the curve with equation $y = 1/x$. Give each answer in form $a + b\sqrt{c}$. [6 marks]

6. Solve simultaneous equations: $\lg(x + 2y) = 0$ and $x^2 + 4xy + y = 1$ [5 marks]

7. Solve simultaneous equations: $x/(2y) - 4y/x = -1$ and $x = 1 - 6y$ [5 marks]

Answer key

1. $x = \pm 1, \pm 2; y = \mp 4, \mp 2$

2. $x=4, y=16$

3. $P=(\sqrt{6}, \frac{2}{\sqrt{6}}), Q=(-\sqrt{6}, -\frac{2}{\sqrt{6}})$

4. $x = \sqrt{2}, y=3\sqrt{2}$ or $x=-\sqrt{2}, y=-3\sqrt{2}$

5. $x = -10 \pm \sqrt{17}, y = \frac{-10 \pm \sqrt{17}}{83}$

6. $y = 0, x = 1$ and $y = \frac{1}{4}, x = \frac{1}{2}$

7. $y = \frac{1}{2}, x = -2; y = \frac{1}{8}, x = \frac{1}{4}$